

CI-1 · RP2354B PoE module

SPECS

Dimensions	96.7mm × 35.6mm
Operating temp.	-20C to +70C
Power output	12 V / 2 A (XT30U-M socket, PoE-supplied) 3.3 V / 1 A (2x 2mm JST sockets, supplied by PoE and USB) 5 V / 4 A (USB/XT30; USB host mode, PoE-supplied)
Ethernet	10/100 Mbit/s PoE IEEE 802.3at class 4
Connectors	40-pin 0.5 mm I/O FFC 9-pin 2.54 mm I/O header (pins shared with FFC connector) 3-pin 1.27mm SWD header USB-C RJ45
Programming	SWD USB-C (in peripheral mode)
Buttons	RESET, RUN
LEDs	Link activity Link speed Power 1× MCU-controlled (GPIO27, shared with FFC connector)
Mounting holes	4x M2 (3.0mm, 3.3mm), (3.0mm, 32.3mm) (94.0mm, 3.3mm), (94.0mm, 32.3mm)

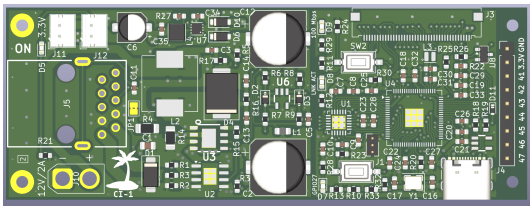
CONTACT INFO

Visit our website for more information, dedicated software, and similar designs.



<http://copperislands.net>
hardware@copperislands.net

CI-1 · RP2354B PoE module · 100Mbit/s · 25W · USB-C host/peripheral



DESCRIPTION

CI-1 is a module intended for use as part of Ethernet devices that would otherwise require a separate PSU. It includes a performant, PIO-enabled RP2354B MCU providing ample flash memory, and interfacing capability for your designs.

FEATURES

- Up to 100Mbit/s Ethernet throughput (LAN8720AI PHY)
- 25W combined power output
- High I/O count: 36x GPIO, 8x ADC
- Optional high-precision LM4040 ADC voltage reference
- USB-C peripheral, and host modes (selectable with configuration strap)
- High-performance MCU: 150 MHz (Arm Cortex-M33/Hazard3 RISC-V), 520 KB SRAM, 2MB Flash, overclocking support
- Long-lifecycle MCU: RP2354B is expected to stay in production until at least January 2045

APPLICATIONS

- Cloud connectivity
- Smart grid
- Automation
- Access control
- Digital signage
- Continuous integration
- USB/IP
- PoS
- AIoT

CI-1 · RP2354B PoE module

BUTTONS

- **RESET (SW1)** — hard reset RP2350 (located near USB, SWD connectors).
- **BOOT (SW2)** — hold when plugging USB cable to enter bootloader (located near FFC connector).

PROGRAMMING

The device can be programmed using the same workflows as a Pico 2 development board, including USB drag-and-drop via the UF2 bootloader, picotool, filesystem-based access, and SWD debugging/programming. Remember to disconnect Ethernet power while programming over USB. USB programming is not available while in USB host mode.

ETHERNET PHY

The LAN8720AI PHY interfaces with the MCU over the RMII bus for Ethernet communication. The following table maps MCU pins to PHY signals:

GPIO	Signal	GPIO	Signal	GPIO	Signal	GPIO	Signal
4	TXD0	7	NRST	10	MDIO	13	RXD0
5	TXD1	8	NINT	11	CRS_DV	14	RXD1
6	TXEN	9	MDC	12	RXER	15	CLKIN

FFC CONNECTOR

The high pin-count, double-sided FFC connector provides a compact interface for connecting modules with a large number of I/O in a small footprint.

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
1	GND	9	GPIO39	17	GPIO47_ADC7	25	GPIO28	33	GPIO3
2	GND	10	GPIO40_ADC0	18	GPIO38	26	GPIO26	34	GPIO22
3	3V3	11	GPIO41_ADC1	19	GPIO35	27	GPIO25	35	GPIO21
4	3V3	12	GPIO42_ADC2	20	GPIO33	28	GPIO24	36	GPIO20
5	GPIO27	13	GPIO43_ADC3	21	GPIO32	29	GPIO23	37	GPIO19
6	GPIO34	14	GPIO44_ADC4	22	GPIO31	30	GPIO0	38	GPIO18
7	GPIO36	15	GPIO45_ADC5	23	GPIO30	31	GPIO1	39	GPIO17
8	GPIO37	16	GPIO46_ADC6	24	GPIO29	32	GPIO2	40	GPIO16

Handle with care

FFC connectors are delicate and designed for limited reuse. Make sure the latch is open and the cable is properly aligned before insertion, then close the lock gently and avoid bending or pulling the cable near the connector.

CI-1 · RP2354B PoE module

SIDE PINS

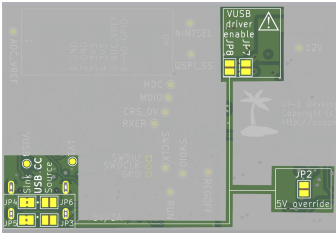
The 9-pin side header provides convenient 3.3V, GND, and GPIO access for breadboard prototyping and quick external circuit connections.

Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal	Pin	Signal
1	GND	3	GPIO41_ADC1	5	GPIO43_ADC3	7	GPIO45_ADC5	9	GPIO47_ADC7
2	3V3	4	GPIO42_ADC2	6	GPIO44_ADC4	8	GPIO46_ADC6		

USB

This device is configured to operate as a USB peripheral by default. To enable host mode:

1. Cut both **sink USB CC** pads (on the back side, next to the USB-C socket).
2. Bridge both **source USB CC** pads.
3. Bridge **5V override** (change PoE regulator output voltage from 12 V to 5 V required to power USB peripherals).
4. Bridge **VUSB driver enable** (connect regulator output to VUSB).
5. Ensure VUSB is held at 5V before connecting USB peripherals.

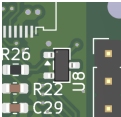


No galvanic isolation

This design does not offer galvanic isolation between USB and PoE ground. Connecting more than one power source at a time may create an unintended current path. To protect your hardware, do not attempt to use this device as a USB peripheral while Ethernet-powered (including while programming over USB).

ADC VOLTAGE REFERENCE

An LM4040 precision reference voltage regulator (U8, SOT-23) can be soldered onto the board to provide a stable, low-noise 3.0 V reference for improving ADC measurement accuracy. This can significantly reduce drift and supply-related variation.



ERRATA

The ADC_VREF signal is incorrectly included in the FFC connector pinout printed on the back-side silkscreen; this signal is not routed to the FFC connector and should be disregarded.